SECTION A-46 MARKS
ATTEMPT ALL QUESTIONS IN THIS SECTION.

1. (a). Define the term **first electron affinity**. (01 mark)

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(b). Explain why the **first electron affinity** is usually **exothermic** while the **second electron affinity** is usually **endothermic**. (02½ marks)

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(c).State any **two factors** that affect the **magnitude of electron affinity**. (01 mark)

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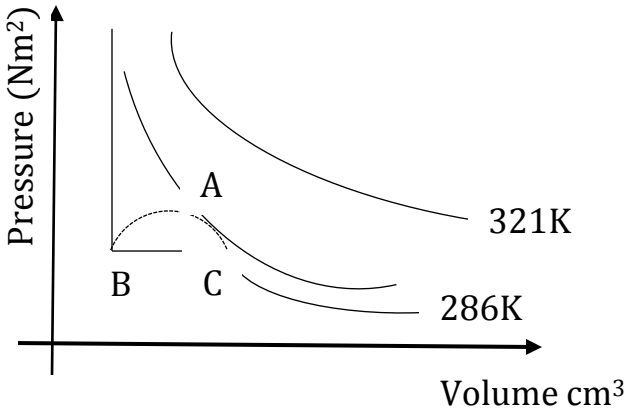
2. (a)The **diagram below** shows isotherms of a gas.



i. What is the **critical temperature** of the gas? (0½ mark)

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ii. Which **isotherm represents** the behaviour of an ideal gas? (0½ mark)

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iii. What does the **region ABC represent**? Give **a reason**. (01 mark)

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iv. Explain **the shape of the curve BE**. (02 marks)

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(b).State **one conditions** necessary for **liquefying a gas**. (01 mark)

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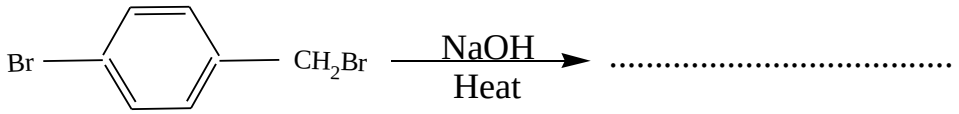
3. (a).An **organic compound X** reacts with hot aqueous sodium hydroxide solution to give **compound W**. complete the **equation of reaction**.

(01 mark)



(b).Name the **type of reaction mechanism** for the reaction of **X** with hot aqueous sodium hydroxide solution. (0½ mark)

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(d).Explain why **W still contains one bromine atom**. (02½ marks)

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4. Aluminium, silicon and phosphorous belong to **period 3** of the periodic table.

(a).For each element, name the **structure** and write the **formula of the oxide**. (03½ marks)

Elements	Aluminium	Silicon	Phosphorous
Formula of oxide			
Structure of oxide			

(b).Write **equation** for the reaction between oxide of silicon and sodium hydroxide solution. (01 mark)

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5. (a).State **Graham's law of gaseous diffusion**. (02 marks)

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(b).A sample of ammonia was placed at **one end A** of the **0.8m** glass tube held horizontally. At the other **end B** of the tube was placed a sample of hydrochloric acid and both ends of the tube closed. When the tube was left for some time, a white ring was formed inside the tube.

Calculate the **distance between end B** and the **white ring**. (03 marks)

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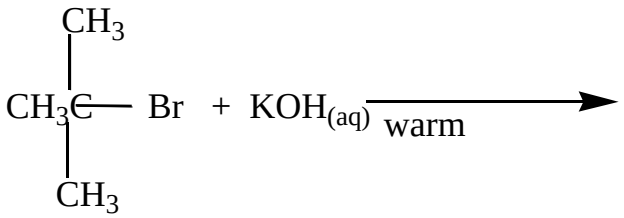
6. (a).Complete the **following equations** below. (@01 mark)

(i). $CH_3CH_2I + KOH_{(aq)} \xrightarrow[\text{warm}]{}$ ……

(ii). 

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(b).Name the **type of reaction mechanism** in both cases. (01 mark)

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(c).Draw a well labelled **diagram of energy** versus **reaction path** for the reaction in (a). (ii). above. (02 marks)

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7. The table below shows the first five successive ionization energies of elements V, **W**, **X**, **Y** & **Z**.

Elements	Ionization energies (KJ/mol)			
	1st	2nd	3rd	4th
V	900	1800	14800	2100
W	500	4600	6900	9500
X	630	1600	3000	4800
Y	740	1500	7700	10500
Z	580	1800	2700	11600

a) What is meant by the term **ionization energy**? (01 mark)

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b) (i).State the element which is most likely to **form ion** with a charge of **+1**.Give a **reason** for your answer. (01½ marks)

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(ii).Identify **two elements** that are in the **same group** of the periodic table. (01 mark)

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(iii).Write the formula of the **chloride of V**. (01 mark)

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8. (a).(i).Define the **radioactivity**. (01 mark)

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(ii).State one property of **gamma radiation**. (01 mark)

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(iii).Give the **effect of beta radiation** to the nucleus of the decaying particle. (01 mark)

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(b).Complete the following **equations for nuclear reactions**. (@01 mark)

i. $^{236}_{92}U \longrightarrow ^{92}_{36}Kr + \ldots\ldots\ldots\ldots + ^{141}_{56}Ba$

ii. $^{239}_{94}Pu + ^{4}_{2}He \longrightarrow \ldots\ldots\ldots\ldots + ^{27}_{13}Al$

9. Explain the following **observations**.

a) Concentrated sulphuric acid is not used in preparation alkyliodide in the presence of hydrogen iodide. (03½ marks)

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b) It is easier to nitrate methylbenzene than benzene itself. (05 marks)

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SECTION B-54 MARKS

ATTEMPT ANY SIX QUESTIONS IN THIS SECTION.

10.(a).State the **conditions** under which each of the following elements reacts with water. In each case, write the **equations for the reaction**.

i. Silicon (01½ marks)

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ii. Chlorine (01½ marks)

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iii. Sodium (01½ marks)

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iv. Magnesium (03 marks)

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(b).Write an **equation** for the reaction between aluminium and sodium hydroxide solution. (01½ marks)

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11.(a).(i).What is the **relationship of Boyle's law** to kinetic molecular theory of gases? (02 marks)

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(ii).Sulphur trioxide and ammonia gases occupy some volumes **at 298K** and **$6.99 \times 10^4 Nm^{-2}$**.Show which gases have the largest volume. (03 marks)

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(b).(i).What is **Dalton's law of partial pressure**? (01 mark)

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(ii).If **20g** of carbon dioxide and **5g** of carbon monoxide gases were put in to a **1.00litre** container at **27°C**.what is the **total pressure** in the container? (03 marks)

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12.A gas **Q** contains **30.43%** nitrogen and the rest being oxygen gas.**0.23g** of **Q** occupied **154.11cm³** at **150°C** and **840mmHg**.

Determine the:

i. Empirical formula of **gas Q**. (02 marks)

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ii. Molecular formula of **gas Q**. (02½ marks)

[**1 mole** of a gas occupies **24dm³** at a temperature of **25°C** and pressure of **760mmHg**]

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(b). When a mixture of **Q** and oxygen was bubbled through water, a compound **Y** was formed.

i. Identify **Y**. (0½ mark)

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ii. Write **equation** for the reaction leading to the formation of **Y**. (01 mark)

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iii. Briefly describe a test(s) that can be carried out to identify the anion in **Y and write equation of the reaction**. (03 marks)

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13. When thin metal foil is subjected to alpha particles from radiation, few alpha particles passed through undeflected.

(a). Explain this observation. (03 marks)

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(b).Mass spectrometer is used to identify isotopes of certain elements.

i. What are isotopes? (02 marks)

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ii. State the function of magnetic field in mass spectrometer. (01 mark)

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iii. Give one advantage of mass spectrometer. (01 mark)

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(c).An element Z has four naturally occurring isotopes with isotopic masses and relative abundance as shown below.

Isotopic mass	Relative abundance (%)
204	01.42
206	24.07
207	22.12
208	52.40

Calculate the average atomic mass of Z. (02 marks)

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14. Complete the following equations and in each case outline a suitable **mechanism** for the reaction.

a. $CH_3CH_2C{\equiv}CH \xrightarrow{2HBr}$ …… (03 marks)

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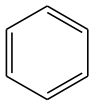
b.  + Conc.H_2SO_4 $\xrightarrow{\text{Conc.}HNO_3}$ ..... (02 ½ marks)

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c. $C(CH_3)_3Br$ $\xrightarrow[\text{Heat}]{NaOH_{(aq)}/CH_3CH_2OH}$ ..... (03½ marks)

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15.The table shows the atomic radius and first ionization energy of some elements in period 3 of the periodic table.

Elements	Na	Mg	Al	Si	P	S	Cl
Atomic radius	0.186	0.160	0.143	0.117	0.110	0.104	0.099
1st I.E(KJ/mol)	496	738	577	787	1060	1000	1251

a. (i).State how atomic radius of the elements varies across the period. (01 mark)

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(ii).Explain your answer in a (i). (03 marks)

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b. (i).Explain how atomic radius affects the ionization energy. (02 marks)

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(ii).Why the first ionization energy of aluminium is lower than that of magnesium. (03 marks)

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THE PERIODIC TABLE

1	2											3	4	5	6	7	8
1.0 H 1																1.0 H 1	4.0 He 2
6.9 Li 3	9.0 Be 4											10.8 B 5	12.0 C 6	14.0 N 7	16.0 O 8	19.0 F 9	20.2 Ne 10
23.0 Na 11	24.3 Mg 12											27.0 Al 13	28.1 Si 14	31.0 P 15	32.1 S 16	35.4 Cl 17	40.0 Ar 18
39.1 K 19	40.1 Ca 20	45.0 Sc 21	47.9 Ti 22	50.9 V 23	52.0 Cr 24	54.9 Mn 25	55.8 Fe 26	58.9 Co 27	58.7 Ni 28	63.5 Cu 29	65.7 Zn 30	69.7 Ga 31	72.6 Ge 32	74.9 As 33	79.0 Se 34	79.9 Br 35	83.8 Kr 36
85.5 Rb 37	87.6 Sr 38	88.9 Y 39	91.2 Zr 40	92.9 Nb 41	95.9 Mo 42	98.9 Tc 43	101 Ru 44	103 Rh 45	106 Pd 46	108 Ag 47	112 Cd 48	115 In 49	119 Sn 50	122 Sb 51	128 Te 52	127 I 53	131 Xe 54
133 Cs 55	137 Ba 56	139 La 57	178 Hf 72	181 Ta 73	184 W 74	186 Re 75	190 Os 76	192 Ir 77	195 Pt 78	197 Au 79	201 Hg 80	204 TI 81	207 Pb 82	209 Bi 83	209 Po 84	210 At 85	222 Rn 86
223 Fr 87	226 Ra 88	227 Ac 89															

139 La 57	140 Ce 58	141 Pr 59	144 Nd 60	147 Pm 61	150 Sm 62	152 Eu 63	157 Gd 64	159 Tb 65	162 Dy 66	165 Ho 67	167 Er 68	169 Tm 69	173 Yb 70	175 Lu 71
227 Ac 89	232 Th 90	231 Pa 91	238 U 92	237 Np 93	244 Pu 94	243 Am 95	247 Cm 96	247 Bk 97	251 Cf 98	254 Es 99	257 Fm 100	256 Md 101	254 No 102	260 Lw 103

♥ ===END===

WELCOME TO SENIOR FIVE, YEAR 2019

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